Wei-Tao Lyu

POSTDOCTORAL FELLOW · DEPARTMENT OF PHYSICS · CUHK

Astronomical instrumentation researcher working on superconducting Microwave Kinetic Inductance Detectors, submillimetre camera systems, cryogenic integration, and readout characterization.

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01 RESEARCH EXPERTISE

MKID Design
Design and characterization of Microwave Kinetic Inductance Detectors for astronomical instrumentation.

Readout Systems
Development and optimization of readout systems for superconducting resonator arrays.

Cryogenic Integration
Cryostat system integration with superconducting MKIDs and cold optics for submillimetre cameras.

02 EDUCATION & APPOINTMENTS

2022 — PRESENT
Postdoctoral Fellow
Department of Physics, The Chinese University of Hong Kong
Supervisor: Prof. Hua-Bai Li

2022
Research Associate
Department of Physics, The Chinese University of Hong Kong
Supervisor: Prof. Hua-Bai Li

2015 — 2021
Ph.D. in Astronomy
School of Astronomy and Space Science, University of Science and Technology of China
Advisor: Prof. Sheng-Cai Shi

2011 — 2015
B.Sc. in Physics
School of Physics, Huazhong University of Science and Technology

03 SELECTED PUBLICATIONS

01 **Laser-Based Frequency-to-Pixel Mapping Method for Submillimeter MKID Array**
Lyu, W., Li, H. B., Zhi, Q., Li, J., & Shi, S. (2026). *RAS Techniques and Instruments*, rzag007.

02 **新一代亚毫米波偏振仪 Roger**
吕伟涛，黄俊锟，孙�淋，李华白 (2025). *天文学进展*, 第43卷 第4期.

03 **ROGer: Remote Observing from Greenland**
Lyu, W., Huang, J., & Li, H.-B. (2024). *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XII*, SPIE 131021D.

04 **Disorder effects in NbTiN superconducting resonators**
Lyu, W. T., Zhi, Q., Hu, J., Li, J., & Shi, S. C. (2024). *Chinese Physics B*, 33(2), 027401.

05 **Cavity and ground effects on a superconducting microstrip resonator**

Lv, W. T., Ding, J. Q., Zhi, Q., Wang, Z., Li, J., & Shi, S. C. (2021). *AIP Advances*, 11(9), 095308.

04 RESEARCH PROJECT

Hong Kong's Window on the Universe

Building a pioneering submillimetre astronomical camera. Role: cryostat system integration with superconducting MKIDs and cold optics.

Submillimetre camera · Superconducting detectors · Cold optics

05 TECHNICAL SKILLS

INSTRUMENTATION

Superconducting resonators, MKIDs, device characterization.

CRYOGENICS

Cryostat integration and optimization for astronomy systems.

DATA ANALYSIS

Signal processing and calibration for superconducting devices.

SOFTWARE

Simulation and data analysis tools for instrumentation work.